DOOM OS · SP-404MKII

Pocket cheat sheet & panel map – a graphical sequencer layer for the Roland SP-404MKII



HOLD this key · PRESS / TAP this key · TURN this knob · ❶ order of steps

orange bar = Roland's printed SHIFT function for that pad

WHERE THINGS LIVE

- **Display + FX row** – DOOM OS draws its grid on the display. FX buttons (FILTER+DRIVE, RESONATOR, DELAY, ISOLATOR, DJFX LOOPER, MFX) flank it.
- **Knobs** – CTRL 1 / 2 / 3 edit whatever the screen shows; VALUE scrolls (turn) and confirms (press).
- **Left block** – PATTERN SELECT, PATTERN EDIT, RECORD SETTING / DEL, REC, RESAMPLE / EXIT, COPY, REMAIN.
- **Right block** – START/END, PITCH/SPEED, MARK / BPM SYNC, GATE, LOOP, REVERSE, ROLL / bank buttons A/F–E/J and SHIFT.
- **Pads** – 16 pads in a 4×4 grid, numbered left→right, top→bottom. In DOOM OS they are the **16 steps**. BUS FX, HOLD, EXT SOURCE and SUB PAD sit in the column on the right.

GOLDEN RULES

- **Open**: hold `REMAIN` + press `SUB PAD`. **Close**: `EXIT` (saves).
- **Lost?** Hold `SHIFT` for 3 s, then press any control to see what it does. `EXIT` leaves help.
- `EXIT` always backs out one level: closes menus, cancels pattern tools, bounces and parameter-lock edits.
- **Undo / redo**: `SHIFT` + `PATTERN SELECT` / `REMAIN` + `PATTERN SELECT`.
- **Saving is automatic** after edits, on pattern change and on exit.
- Combos are written **hold + press**: the first key is held down, the second is tapped.

BOOKLET MAP

p2 Getting around
open · play · patterns · tempo · tracks · steps · menus · grid · piano roll

p3 Record & edit
rec modes · quantize · bounce · pattern tools · parameter locks

p4 Sound & mix
instruments · kits · arp · mixer · EQ · side chain

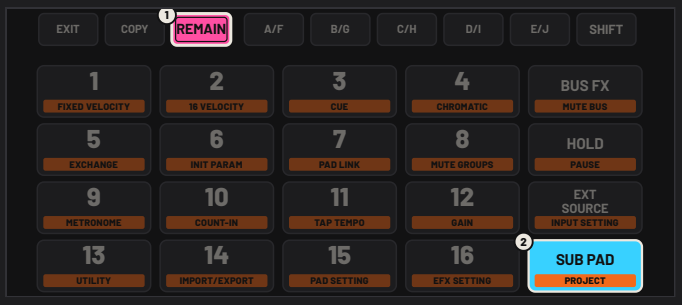
p5 FX & reference
FX · automation · settings · master shortcut table

Unofficial cheat sheet compiled from the DOOM OS 0.7 manual (Klangfeld Labs) with panel layout taken from Roland's SP-404MK2 Reference Manual. Not affiliated with Roland. Diagrams are schematic, not to scale.

Getting around

Open · play · patterns · tempo · tracks · steps · menus · grid · piano roll

Open / close DOOM OS



Open: hold **REMAIN** + press **SUB PAD**.
Close: press **EXIT** – saves changes. A *Hold to Exit* setting can require holding it.

Play / stop · undo · redo



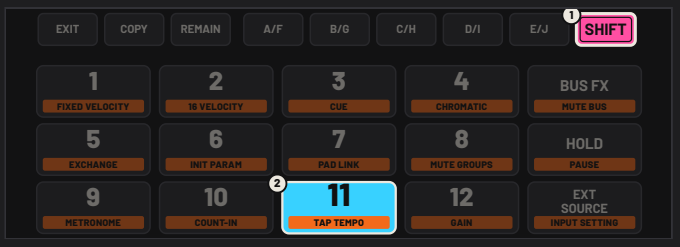
Play / stop: **PATTERN SELECT**
Undo: hold **SHIFT** + **PATTERN SELECT**
Redo: hold **REMAIN** + **PATTERN SELECT**

New / switch pattern



New: hold **PATTERN SELECT** + an **empty** pad.
Switch: hold **PATTERN SELECT** + a pad – starts when the playing pattern ends. **Add** **SUB PAD** to switch immediately.

Tempo



SHIFT + **pad 11** (TAP TEMPO), or press **RECORD SETTING** twice > BPM.

Add a track



Sample track: hold **SUB PAD**, tap sample pads, release.
Instrument track: hold **SUB PAD** + tap an **empty** pad (opens the instrument browser, p4).

Steps & notes (CTRL 1 / 2 / 3)



The 16 pads are the 16 **steps**: tap to toggle, hold a pad to select that note.
CTRL 1 velocity · **CTRL 2** length · **CTRL 3** nudge. Turn with **no pad held** = whole track; hold a pad = that one note.

Do this	Input	Result
Fine edits	**SHIFT** + **CTRL 2** · **SHIFT** + **CTRL 1**	Length in 1/128 steps · pitch up to ±12 semitones. Keep a selection: hold pad + **REMAIN**
Mute / solo track	**SHIFT** + press **VALUE** · **SHIFT** + **ROLL**	
Copy / paste / delete	**COPY**, then **RECORD SETTING** > EDIT > PASTE · **DEL**	DEL asks first; CLEAR... asks YES/NO; PASTE does not ask
TRACK page	**RECORD SETTING** once	EDIT (SWAP PAD, PASTE, CLEAR..., BOUNCE...) · PATTERN (CLEAR..., BOUNCE...) · GENERATE (PRINT ARP..., CLEAR...) · VIEWS (MIXER, AUTOMATION, ARP, SIDECHAIN)
PATTERN panel	**RECORD SETTING** twice (3rd closes)	BPM · LENGTH (bars) · REC mode

Patterns over 512 notes, and effect movements, can't be undone.

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Record & edit

Recording modes · quantize · bounce in place · pattern tools · parameter locks

Record (REC)
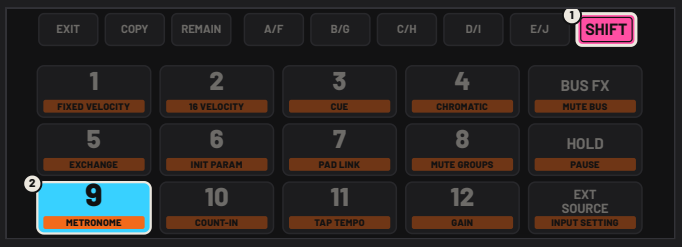


1st press = **rehearsal** (nothing recorded). 2nd press = **records**. EXIT stops recording (pattern keeps playing). HOLD is ignored while REC is on.

Record mode



SHIFT + REC cycles **SINGLE** · **MULTI** · **CHROMATIC** (pad 5 = original pitch) · **16 VEL** (pad 4 = 127, pad 13 = 1). Also set in the PATTERN panel.

Metronome



SHIFT + pad 9 cycles ON · RECPLAY · OFF.

Input quantize (new notes)



SHIFT + PATTERN EDIT sets **QNT** · **SWG** · **VEL** · **LEN** for newly recorded notes.

Bounce in place



RESAMPLE = bounce the **track**; SHIFT + RESAMPLE = **everything you hear**. Pick type with VALUE, tap an **empty pad**, then press RESAMPLE or VALUE. A track bounce swaps the track's notes for one note on the new pad – SHIFT + PATTERN SELECT undoes it; EXIT cancels.

Pattern tools



PATTERN EDIT > turn VALUE to a tool > press VALUE > set amount > press VALUE to apply. Tap a track pad to limit to that track; tap again for ALL.

Parameter locks



MARK (grid or editor) > tap a step > edit with CTRL 1-3 > MARK saves, EXIT discards. Locked pads show **blue**; DEL resets a step.

Pattern tool	Range / note
QUANT	1/4 to 1/32, incl. triplets
SWING	50–75; SHIFT switches 8th / 16th
SHIFT	±16 steps, wraps around
DOUBLE · HALVE	DOUBLE max 64 bars / 512 notes; HALVE not on a 1-bar pattern
REVERSE	–
HUMAN · TRANSP	HUMAN 1–30 · TRANSP ±12 semitones

Bounce type	What it captures
CUT	Pattern length
TAIL	Includes effect decay, up to 30 s
LOOP	Loops from the second pass

PARAMETER-LOCK FACTS

- Per-step instrument parameters, instrument tracks only.
- Locks follow notes when moved (SHIFT tool, QUANTIZE, DUPLICATE, CROP, SWAP PAD, COPY, PASTE).
- Limits: 4096 locks per project · 64 steps per session · 126 locks per track in a pattern.

View	Input	Result
Grid views	SHIFT + RECORD SETTING	Cycles GRID (8 tracks) · LABELS · BEATS · BOTH · ZOOM (4 larger tracks)
Zoom · move · follow	REMAIN +turn VALUE · SHIFT +turn VALUE · SHIFT + HOLD	Zoom in ZOOM view · move by bar (beat at 1-beat zoom) · toggle Follow (playhead)

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Sound & mix

Instruments · drum kits · arpeggiator · mixer · EQ · side chain · waveform

Instrument browser



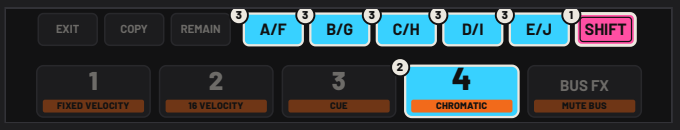
Hold **SUB PAD** + tap an **empty** pad, release. Turn **VALUE** to choose; hold **SUB PAD** to preview; **VALUE** loads. 20 instruments and 11 drum kits are built in.

Instrument editor



On an instrument track press **START/END**. Turn **VALUE** to pick a section (TONE, MOD, AMP, OUT…); **CTRL 1-3** edit; **SHIFT** = finer steps; **EXIT** saves.

Chromatic keys & octave



In the editor: **SHIFT** + **pad 4**. Octave via the BANK buttons: **C/H** default · **B/G**, **A/F** down · **D/I**, **E/J** up.

Drum kits



SHIFT + turn **VALUE** to KITS, choose, press **VALUE**, pick a bank with a BANK button, press **VALUE**. A non-empty bank asks to overwrite (can't undo); protected banks are refused.

Mixer



Hold **RECORD SETTING** 0.6 s. **VALUE** selects a track; **CTRL 1** pan · **CTRL 2** bus (B1 · B2 · DRY) · **CTRL 3** level. Tap a pad = mute; **REMAIN** + pad = solo. **VALUE** cycles pan / bus / level for fine adjust; **EXIT** deselects.

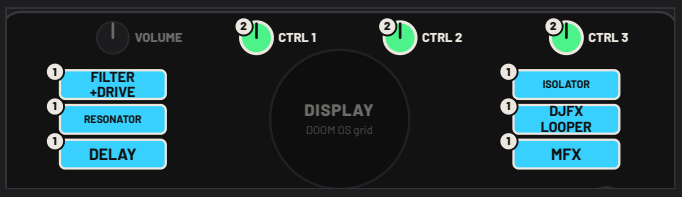
Tool	How	Details
Arpeggiator	**RECORD SETTING** > ARP, or the grid while playing	Hold pads to arp. **VALUE** toggles arp and swaps PLAY / PITCH pages; **HOLD** keeps it running. Persists until power-off. **PLAY**: **CTRL 1** TYPE (UP, DOWN, U/D, RND, ORD, CHRD, CONV, DIV, PNKY, THMB, U/D+) · **CTRL 2** RATE 1/4–1/32 + triplets · **CTRL 3** GATE 1–100%. **PITCH**: KEY (C–B) · SCALE (40) · OCT (±3). Print: **RECORD SETTING** > GENERATE > PRINT ARP… (replaces all notes on the track)
Track EQ	in mixer, **PITCH/SPEED**	**CTRL 1** gain ±12 dB · **CTRL 2** freq 20 Hz–20 kHz · **CTRL 3** Q 0.3–10. **VALUE** opens the screen; again = ADV band select. Up to 16 pads; EQ belongs to the pad
Side chain	**RECORD SETTING** > SIDECHAIN	**VALUE** on a **source** bus, then a **target** bus; **VALUE** twice on one bus = compress it. **CTRL 1** THR · **CTRL 2** RAT · **CTRL 3** REL; GAIN via **VALUE**. Off: hold **DEL** + **VALUE** on the bus. Shared with the unit's own side chain, all projects
Piano roll	press **VALUE** on the grid	Hold a step's pad + press another pad to set pitch (pad 5 = original; +1 semitone per pad across, and up the rows). Turn **VALUE** scrolls; **SHIFT** +turn changes track
Waveform display	**SHIFT** + **START/END**	1× output · 2× selected track · 3× hide. Shows the last two seconds
Replace instrument	in editor: **REMAIN**, choose, YES	Notes stay, parameters reset
Remove instrument	outside DOOM OS: hold **DEL** + pad, then **VALUE**	Cannot be undone
Mixer display	**SHIFT** + **RECORD SETTING**	ARROWS / DITHER, with or without NAMES

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FX & quick reference

Effects · automation · settings · every shortcut on one page

Effects



Each FX button turns its effect **on** and opens its page (stays open until **EXIT**; the effect stays on). Press again to turn **off**. **CTRL 1-3** set parameters; hold **VALUE** for page 2. **REMAIN** + FX button reopens a page without toggling.

Bus FX settings · record movements



SHIFT + **pad 16** opens bus effects settings. Record movements: open an FX page, press **REC** twice, turn knobs. **BUS FX** switches buses.

FX automation



SHIFT + **MARK** opens it. Turn **VALUE** for a parameter; **SHIFT** + turn = next effect. Press **VALUE** for per-step view, tap a pad, turn **VALUE**. **START/END** toggles **EFX MOTION PLAY** – movements only play while it is on.

MASTER SHORTCUT TABLE

Function	Input	Function	Input
Open DOOM OS	**REMAIN** + **SUB PAD**	Bounce track / mix	**RESAMPLE** / **SHIFT** + **RESAMPLE**
Undo / Redo	**SHIFT** + **PATTERN SELECT** / **REMAIN** + **PATTERN SELECT**	Instrument editor	**START/END**
Mixer	hold **RECORD SETTING** 0.6 s	Parameter locks	**MARK**
Help mode	hold **SHIFT** 3 s	FX automation	**SHIFT** + **MARK**
Metronome	**SHIFT** + **pad 9**	Waveform display	**SHIFT** + **START/END**
Record mode	**SHIFT** + **REC**	Bus FX settings	**SHIFT** + **pad 16**
Tempo	**SHIFT** + **pad 11**	UTILITY menu	**REMAIN** + **PATTERN EDIT**
MFX effect select	hold **MFX** + pad (17–32 via **DJFX LOOPER**; 33–48 via **ISOLATOR**)	Piano roll	press **VALUE** on the grid
Input quantize	**SHIFT** + **PATTERN EDIT**	DOOM OS settings	**EXIT**, then **SHIFT** + **pad 13** > SYSTEM > DOOM OS

PAD SHIFT FUNCTIONS

1 FIXED VELOCITY	2 16 VELOCITY	3 CUE	4 CHROMATIC
5 EXCHANGE	6 INIT PARAM	7 PAD LINK	8 MUTE GROUPS
9 METRONOME	10 COUNT-IN	11 TAP TEMPO	12 GAIN
13 UTILITY	14 IMPORT/EXPORT	15 PAD SETTING	16 EFX SETTING

Underlined = pad shortcuts DOOM OS refers to. Labels are Roland's printed SHIFT functions.

DOOM OS SETTINGS

Path: **EXIT**, then **SHIFT** + **pad 13** > SYSTEM > **DOOM OS** tab (**CTRL 3** navigates). Options: Mixer Level (ARROWS / DITHER) · Mixer Names · Boot Anim (GRAFFITI, VHS, HELLO, BUBBLE, SKETCH, HYPERSPACE, RANDOM) · Pattern View · Grid Lines · Pattern Follow · Motion Rec (OVERWRITE / OVERDUB) · Hold to Exit (0.1–1.0 s or OFF) · Boot to DOOM (YES / NO). The SHORTCUT tab holds up to 8 custom shortcuts.

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